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1
2 // COS30008, Problem Set 4, Problem 2, 2022
3
4 #pragma once
5
6 // #include "BinarySearchTreeIterator.h"
7 #include "BinaryTreeNode.h"
8
9 #include <stdexcept>
10
11 // Problem 3 requirement
12 template <typename T> class BinarySearchTreeIterator;
13
14 template <typename T> class BinarySearchTree {
15     private:
16         using BNode = BinaryTreeNode<T>;
17         using BTreeNode = BNode *;
18
19         BTreeNode fRoot;
20
21     public:
22         BinarySearchTree()
23             : fRoot(&BNode::NIL) {
24
25         };
26
27         ~BinarySearchTree() {
28             if (!this->fRoot->empty()) {
29                 delete this->fRoot;
30             }
31         };
32
33         bool empty() const { return this->fRoot->empty(); };
34         size_t height() const { return this->fRoot->height(); };
35
36         bool insert(const T &aKey) {
37             if (this->empty()) {
38                 this->fRoot = new BinaryTreeNode<T>(aKey);
39                 return true;
40             }
41             return this->fRoot->insert(aKey);
42         }
43         bool remove(const T &aKey) {
44             if (this->empty()) {
45                 return false;
46             }
47             return this->fRoot->remove(aKey, this->fRoot);
48         }
49 }
```

```
50     // Problem 3 methods
51
52     using Iterator = BinarySearchTreeIterator<T>;
53
54     // Allow iterator to access private member variables
55     friend class BinarySearchTreeIterator<T>;
56
57     Iterator begin() const { return Iterator(*this).begin(); }
58     Iterator end() const { return Iterator(*this).end(); }
59 };
60
```